Problem 1:

Pf: by induction

$P(n) ::=$ every n-vertex graph with width, w, is $(w+1)$-colorable.

Base case:

$P(1) = w = 0$ colorable = 1

Inductive step:

Assume $P(n)$ is true for induction

Let $G = (V, E)$ be any $(n+1)$ notes, with width, w

1. remove the last vertex, v_{n+1} to get G' with n vertex with width w
2. by $P(n)$, G' is $w+1$ colorable
3. add back v_{n+1} which have w, width
4. since we have $[w+1]$ color but a color from $(w+1)$ that is not color of neigborhood of v_{n+1}

$P(n+1)$ ✓ □ $\forall n \geq 1$, $P(n)$ is True

Problem 2:

(a) Theorem: Any subgraph of a planar graph is planar

Proof:

Let $G = (V, E)$ be any planar graph

Let $G' = (V', E')$ be any subgraph of G, meaing $V_0 \subseteq V$ & $E_0 \subseteq E$

since subgraph G' doesn't add new vertixes and edges,

the resulting of G' has zero edge crossings.

$\therefore G'$ is a planar □

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(2)

Theorem : Any planner graph can be colored with at most color.

Pf: by induction

P(n) ::= Any n-vertex plannar graph can be 6-colored

Base Case:

P(1) = 1 ✓

Inductive step:

for $\forall n \geq 1$, $P(n) \Rightarrow P(n+1)$

Assume P(n) = True & G = ony planner graph with n+1 nodes

take node v with at most degree 5 and remove it from G. to get G' subgraph.

G' is colorable by 6 colors.

Add the v node back with at most degree 5. choose one color that left

P(n+1) ✓

By the principle of induction, $\forall n \geq 1$ P(n) is true.

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